

ITMA 99

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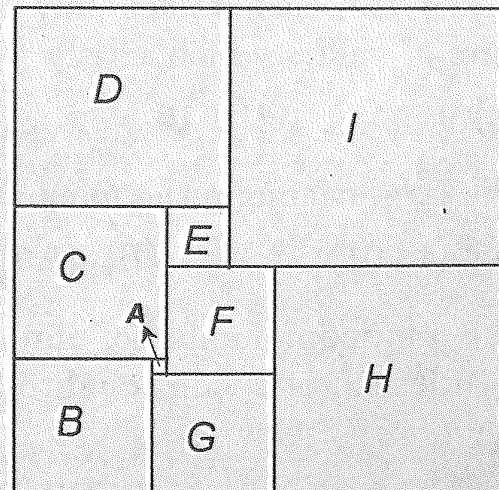
SCREENING TEST - JUNIOR LEVEL

1. If $x^9 = 9^{9^9}$, then x is
(A) 9^9 (B) 9^{8^9} (C) 9^{9^8} (D) 9^{9^9-1} (E) $(9^9)^8$
2. If $1 \times 2 \times 3 \times 4 \times \cdots \times 39 \times 40$ is written in the system with base 13, then the number of zeros with which the resulting number ends is
(A) 3 (B) 0 (C) 4 (D) 2 (E) 8
3. The number of prime numbers p such that $1999! + 1 < p < 1999! + 1999$ is ($n! = 1 \times 2 \times 3 \times \cdots \times n$)
(A) 1 (B) 1998 (C) 1999 (D) 2000 (E) 0
4. Which of the following numbers is different from the others?
(A) $\frac{1996}{1997}$ (B) $\frac{996}{997}$ (C) $\frac{1997996}{1998997}$ (D) $\frac{19971996}{19981997}$ (E) $\frac{996996}{997997}$
5. A secretary has 4 letters addressed to 4 different persons with 4 envelopes bearing the addresses of these persons. The number of ways the careless secretary can put the 4 letters in the envelopes so that no person gets the right letter is
(A) 9 (B) 24 (C) 16 (D) 8 (E) 4
6. A, B, C are three finite sets. A has twice as many elements as B . B has more elements than C . The number of subsets of B is 15 more than that of C . Then the number by which the number of subsets of A exceeds the number of subsets of B is
(A) 0 (B) 2^8 (C) 240 (D) 2^{15} (E) 15
7. If in $\triangle ABC$, $\angle A = 90^\circ$, $\angle B = 15^\circ$ and $BC = 4$ cm, then its area, in cm^2 , is
(A) 16 (B) 8 (C) 4 (D) 2 (E) 1
8. If $x^2 + x + 1 = 0$, then the value of $x^{1999} + x^{2000}$ is
(A) 0 (B) 1 (C) -1 (D) 3999 (E) 2

9. If the common points of the graphs of $y = x^2 - 6x + 2$ and $x + 2y = 4$ are A, B then the midpoint of AB has coordinates
(A) $(-3, 4)$ (B) $(-\frac{3}{4}, \frac{4}{3})$ (C) $(\frac{11}{4}, \frac{5}{8})$ (D) $(0, 4)$ (E) $(4, 0)$
10. The smallest natural number n such that $n!$ (see question 3) is divisible by 1000 is
(A) 20 (B) 15 (C) 10 (D) 12 (E) 25
11. If 1 and the number are also counted among the divisors of a natural number, the number of divisors of the sum of all two digit numbers which leave remainder 1 when divided by 5 is
(A) 0 (B) 2 (C) 3 (D) 4 (E) 6
12. If the radius of a given circle is increased two fold, the ratio of the circumference of the circle so obtained to its diameter is
(A) π (B) 4π (C) $\pi + 2$ (D) $2\pi + 2$ (E) 6π
13. If $[x]$ denotes the largest integer less than or equal to x , then $[\frac{1}{2}] + [\frac{1}{2} + \frac{1}{100}] + [\frac{1}{2} + \frac{2}{100}] + [\frac{1}{2} + \frac{3}{100}] + \dots + [\frac{1}{2} + \frac{61}{100}]$ is
(A) 9 (B) 10 (C) 11 (D) 12 (E) 0
14. If the length of a diagonal of a cube is $\sqrt{12}$ cm, the area of each of its faces, in cm^2 , is
(A) 4 (B) $2\sqrt{3}$ (C) $4\sqrt{3}$ (D) $3\sqrt{2}$ (E) 12
15. The sum of the digits of the number $1000^{20} - 20$ expressed in decimal notation is
(A) 510 (B) 520 (C) 530 (D) 680 (E) 630
16. 3 men and 5 boys can do $\frac{19}{20}$ of a work in 3 days. 4 men and 8 boys can do $\frac{14}{15}$ of the work in two days. How many days does a boy take to finish the whole work?
(A) 25 (B) 28 (C) 30 (D) 32 (E) 35

17. Two clocks are set at the same time, one is seen to gain 40 seconds and the other to lose 50 seconds in 24 hours. In how much time will they show a difference of 15 minutes?
- (A) 90 days (B) 10 days (C) 90 hours (D) 10 hours (E) 15 days
18. The line parallel to the parallel sides of a trapezium passing through the midpoints of the slant sides divides the trapezium in the ratio 5:2. Then the parallel sides are in the ratio
- (A) 13:1 (B) 25:4 (C) 5:2 (D) $\sqrt{5} : \sqrt{2}$ (E) 2:5
19. If x, y are natural numbers, the number of pairs (x, y) for which $x^2 - y^2 = 31$ is
- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4
20. A small group of tribals have a language all of whose words can have only at the most 4 letters while they have only 4 letters in the alphabet of the language. The number of words in the language is
- (A) 256 (B) 320 (C) 336 (D) 340 (E) infinite

21. In the adjoining figure $A, B, C, D, E, F, G, H, I$ are all squares of which A and B have respectively area 1 and 81 cm^2 . Then the area of the square I , in cm^2 , is



- (A) 196 (B) 256
(C) 289 (D) 324
(E) 361

22. Three circles touch each other externally and all the three touch a line. If two of them are equal and the third has radius 4 cm, the radius of the equal circles, in cm, is
(A) 16 (B) 24 (C) 18 (D) 20 (E) 12
23. In a long jump qualifying competition the average jump length of successful competitors was 6.5m; the average jump length of unsuccessful competitors was 4.5m and the average jump length of all the competitors was 4.9m. The percentage of the competitors who were successful was
(A) 4 (B) 16 (C) 20 (D) 25 (E) 55
24. In a given polygon each interior angle at a vertex is $7\frac{1}{2}$ times the exterior angle at the vertex. Then the polygon is
(A) 16 sided (B) 13 sided and not regular (C) 17 sided and not regular
(D) 19 sided and regular (E) 17 sided
25. If $x + y = 5xy$, $y + z = 6yz$, $z + x = 7zx$, then $x + y + z$ is
(A) $\frac{1}{9}$ (B) $\frac{13}{12}$ (C) 9 (D) $\frac{12}{13}$ (E) $\frac{15}{12}$
26. The last two digits of 6^9 are
(A) 96 (B) 16 (C) 36 (D) 56 (E) 76
27. The students in the school-end class in a school exchanged one copy each of photographs of each other. If the total number of photographs which were exchanged was 870, the number of students in the class was
(A) 869 (B) 435 (C) 30 (D) 45 (E) 50
28. If x, y can take values only among natural numbers, the number of such pairs of values which satisfy the equation $2x + 5y = 100$ is
(A) 9 (B) 10 (C) 11 (D) 4 (E) not finite
29. The largest among the numbers $(1.001)^{1000}$, $(1.01)^{1000}$, 1000, 2^9 , 3^6 is
(A) $(1.001)^{1000}$ (B) $(1.01)^{1000}$ (C) 1000 (D) 2^9 (E) 3^6
30. The minimum number of unit squares needed to cover an equilateral triangle with length of a side equal to 2 units is
(A) 5 (B) 4 (C) 6 (D) 9 (E) 3